Papers on systematic review and meta analysis using LLMs

- PLuo X, Chen F, Zhu D, et al. Potential Roles of Large Language Models in the Production of Systematic Reviews and Meta-Analyses. *J Med Internet Res*. 2024;26:e56780. Published 2024 Jun 25. doi:10.2196/56780

- Shen J, Luo Z, Jia D, Wang S, Sun F, Wu J. Large language model enhanced framework for systematic reviews and meta-analyses. *BMJ Digital Health & AI*. 2025;1:e000017. https://doi.org/10.1136/bmjdhai-2025-000017

- Qureshi R, Shaughnessy D, Gill KAR, Robinson KA, Li T, Agai E. Are ChatGPT and large language models "the answer" to bringing us closer to systematic review automation? Syst Rev. Apr 29, 2023;12(1):72. [FREE Full text] [doi:10.1186/s13643-023-02243-z] [Medline: 37120563]

- Li M, Sun J, Tan X. Evaluating the effectiveness of large language models in abstract screening: a comparative analysis. *Syst Rev*. 2024;13(1):219. Published 2024 Aug 21. doi:10.1186/s13643-024-02609-x

- Cao C, Arora R, Cento P, Manta K, Farahani E, Cecere M, et al. Automation of systematic reviews with large language models. *medRxiv*. Published 2025. doi:10.1101/2025.06.13.25329541

- Jaumann C, Wiedholz A, Friedrich A. LGAR: Zero-shot LLM-guided neural ranking for abstract screening in systematic literature reviews. In: Che W, Nabende J, Shutova E, Pilehvar MT, eds. *Findings of the Association for Computational Linguistics: ACL 2025*. Association for Computational Linguistics; 2025:7910-7927. doi:10.18653/v1/2025.findings-acl.412

- Oami T, Okada Y, Nakada TA**.** Accuracy and reliability of data extraction for systematic reviews using large language models: a protocol for a prospective study. *medRxiv*. Published 2024. doi:10.1101/2024.05.22.24307740

- Scherbakov D, Hubig N, Jansari V, Bakumenko A, Lenert LA**.** The emergence of large language models as tools in literature reviews: a large language model-assisted systematic review. *J Am Med Inform Assoc.* 2025;32(6):1071-1086. doi:10.1093/jamia/ocaf063

- Lai, H., Liu, J., Bai, C. *et al.* Language models for data extraction and risk of bias assessment in complementary medicine. *npj Digit. Med.* 8, 74 (2025). https://doi.org/10.1038/s41746-025-01457-w

- Paper: Accelerating clinical evidence synthesis with large language models

PICO Papers

Paper: Automated Mass Extraction of Over 680,000 PICOs from Clinical Study Abstracts Using Generative AI: A Proof-of-Concept Study

Paper: AlpaPICO: Extraction of PICO frames from clinical trial documents using LLMs

Paper: Towards precise PICO extraction from abstracts of randomized controlled trials using a section-specific learning approach

Paper: Extracting PICO Elements From RCT Abstracts Using 1- 2gram Analysis And Multitask Classification

Paper: Not so weak PICO: leveraging weak supervision for participants, interventions, and outcomes recognition for systematic review automation

Paper: Exploring the Use of a Large Language Model for Data Extraction in Systematic Reviews: a Rapid Feasibility Study

Paper: PICOs-RAG: PICO-supported Query Rewriting for Retrieval-Augmented Generation in Evidence-Based Medicine

Paper: Sent2Span: Span Detection for PICO Extraction in the Biomedical Text without Span Annotations

Paper: Improving reference prioritisation with PICO recognition

Paper: Advancing Evidence-Based Medicine for Population, Intervention, Comparison, and Outcome Element Recognition and Extraction in Medical Literature: Large Language Model Approach

Paper: PICO entity extraction for preclinical animal literature

Paper: Unlocking the Power of Deep PICO Extraction: Step-wise Medical NER Identification

Paper: PICO Corpus: A Publicly Available Corpus to Support Automatic Data Extraction from Biomedical Literature

Paper: LLM-UMLS-PICO: A Large Language Model-Based PICO Extraction Method with UMLS Semantic Validation

Paper: A span-based model for extracting overlapping PICO entities from randomized controlled trial publications

Paper: Pretraining to Recognize PICO Elements from Randomized Controlled Trial Literature

Paper: PICO element detection in medical text without metadata: Are first sentences enough?

Paper: Exploring the Use of a Large Language Model for Data Extraction in Systematic Reviews: a Rapid Feasibility Study

Paper: Extracting Health Evidence Information from Biomedical Literature using Large Language Models

Paper: Enhancing systematic reviews in orthodontics: a comparative examination of GPT-3.5 and GPT-4 for generating PICO-based queries with tailored prompts and configurations

Paper: Do it faster with PICOS: Generative AI-Assisted systematic review screening

Paper: Chained Prompting for Better Systematic Review Search Strategies

Paper: Large language models streamline automated systematic review: A preliminary study

Paper: PICOT questions and search strategies formulation: A novel approach using artificial intelligence automation

Paper: A large language model framework for knowledge graph construction of randomized controlled trials for evidence synthesis and querying

Paper: DISTANT-CTO: A Zero Cost, Distantly Supervised Approach to Improve Low-Resource Entity Extraction Using Clinical Trials Literature

Paper: Explainable Prediction of Medical Codes from Clinical Text

Paper: TrialSieve: A Comprehensive Biomedical Information Extraction Framework for PICO, Meta-Analysis, and Drug Repurposing

Datasets:

1. EBM NLP mod dataset: link
2. EBM NLP dataset: link
3. EBMNLP, EBM-COMET,EBM-NLP□□□ and EBM-NLPh: link